

FINITE-SAMPLE SAFE MODEL RANKING UNDER SUBGROUP-MIX TURNOVER

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ABSTRACT

At deployment the class/subgroup mixture of a fixed pool of candidate models can shift (label-prior / subgroup-mix turnover) while each subgroup’s class-conditional risk stays unchanged, so a model optimal for the collected mixture may cease to be optimal—and a worse model can be selected. We formalize *safe ranking* as a finite-sample certification problem: given a calibration split, decide for every future mixture weight w whether to *commit* a model with a certificate that its regret (loss minus the oracle-best model’s loss at w) is at most an operator-set tolerance τ , and otherwise *abstain* honestly rather than hard-select a possibly-overturning pick. We prove a linear-band structure (P1) giving joint coverage $\geq 1 - \delta$ of every mixture risk by weighted Clopper–Pearson bands, and a *paired-difference* (MCB-style) certificate (P3/Thm. 1) that bounds $\max\{0, \max_{j \neq i^*} (R_{i^*} - R_j)\}$ by pairing models on the *same* calibration points, cancelling shared error. On four public carriers and four classifiers, the dependent held-out estimated-regret diagnostic finds no committed row above τ (reported violation fraction 0/committed rows), and naively *minimizing the upper band* is shown to be a vacuous or *anti-selective* selector under heteroskedastic group widths—so we use *point-estimate selection + certificate gating* (M2/M2.5), abstaining exactly on turnover-skew mixtures where forced point-estimate selection has higher observed regret. The paired certificate exposes an explicit *certificate-price frontier*: the exact finite-sample bounds commit 0.269–0.260 (matched 5-seed means); a separately labelled asymptotic normal diagnostic has a 0.503 committed rate, but is not finite-sample evidence. Under a *label budget* (Sec. 6), the dependent held-out diagnostic finds no violation over 3360 static-policy cells; the displayed aggregate commitment rates rank uniform above variance-aware rules on MNIST but below them on Fashion at low budget. These allocation contrasts are descriptive constructed-mixture results, not seed-level uncertainty estimates. Finally, we use a fixed-budget *certificate-width surrogate* (Sec. 7) to diagnose why allocations differ. Its water-filling calculation is not an optimization theorem for the actual candidate- and mixture-indexed regret UCB or a safe allocation rule.

1 INTRODUCTION

Model selection is usually done once, on the mixture where data was collected. But at deployment the label/subgroup mix can move—a model trained on general web traffic now serves a skewed query stream; a classifier tuned on balanced data is deployed onto a single class-dominant population. Under *label-prior shift* (*subgroup-mix turnover*) the class-conditional distribution $P(X \mid Y=g)$ is fixed but the mixture weight w over subgroups is not. A model i ’s risk is then linear in w ,

$$R_i(w) = \sum_g w_g r_{i,g}, \quad r_{i,g} = \mathbb{E}[\ell(\text{pred}_i, Y) \mid Y=g], \quad (1)$$

so the identity of the best model, $\arg \min_i R_i(w)$, can change as w moves across the simplex. Standard selection heuristics live at one of two extremes: *single-point* selection (pick by the collected-mixture estimate) ignores w entirely; *worst-group / DRO / balanced-accuracy* selection ignores where w actually is. None answers the deployment question: *for a given future mixture w , can I both pick a model and certify its regret in finite sample—and when must I abstain rather than risk a pick that overturns $w \mapsto w'$?*

We study this as a *finite-sample safe ranking* problem with abstention: from a held-out calibration split, produce for each w a decision *commit*(model i^* and a certificate that its regret $\leq \tau$) or *abstain* (report that w is undecidable). The certificate is a finite-sample, coverage-guaranteed upper bound on the chosen model’s regret, and is allowed to abstain where the certificate is wider than the operator-set tolerance (Sec. 5, Tab. 2). Abstention is a certificate outcome, not an operational-utility claim: its fallback, latency, and monetary costs are unmeasured here.

1.1 CONTRIBUTIONS

1. **Linear-band ranking structure** (Sec. 3): if each entry $r_{i,g}$ is covered by a Clopper–Pearson interval with cell confidence $\delta/(Gk)$, then every mixture risk satisfies $L_i(w) \leq R_i(w) \leq U_i(w)$ with joint coverage $\geq 1 - \delta$ (P1, Lemma 1); best-vs-rest regret of any chosen model has a *sound* upper bound $U_{i^*}(w) - \min_j L_j(w)$ (P2).
2. **Paired-difference (MCB-style) certificate** (P3, Thm. 1): rather than subtract two *absolute* bands, bound $\max_j(R_{i^*} - R_j)$ by *paired within-CAL* differences (same points, same group). Pairing can reduce shared-error width, but no universal ordering against the absolute bound is claimed.
3. **A mechanism-level caution on “minimize-the-upper-band” selection** (Sec. 4): this pessimistic rule degenerates to the point estimate on homogeneous widths (vacuity) and *can* reverse rank opinion under heteroskedastic widths (Prop. 3); on real data it disagrees with point-estimate selection without a consistent direction. Framed as *bands gate, they do not choose*.
4. **Certificate-price frontier** (Sec. 5, Tab. 3): the exact paired Maurer–Pontil variant commits 0.260 vs. M2’s 0.269 (matched 5-seed) with reported held-out violation fraction 0 among committed rows. The 0.503 normal/CLT result is reported only as an asymptotic diagnostic, never as finite-sample support.
5. **Subgroup label allocation under a budget (M3, Sec. 6)**: static FIT-specified allocations preserve the certificate assumptions. Their constructed-mixture commitment contrasts are reported descriptively, because the frozen M3 summaries do not retain per-seed allocation outcomes for uncertainty analysis.
6. **Fixed-budget width-surrogate diagnostic (M3.5, Sec. 7)**: for a stylized $\beta_g/\sqrt{n_g}$ width proxy, a single fixed mixture has the familiar water-filling solution $n_g \propto (w_g\beta_g)^{2/3}$ on positive coefficients (and zero allocation on zero coefficients). We use this only to interpret same-budget allocation experiments; it does *not* optimize the paired regret UCB, prove safe allocation, or prescribe a deployment gate.

Label	Decision role	Status	Primary location
M1	upper-band selection	negative diagnostic	Sec. 4
M2/P2	estimate + absolute gate	exact certificate	Prop. 1, Tab. 2
M2.5/P3 (MPB/Hoeffding)	estimate + paired gate	exact certificate	Thm. 1, Tab. 3
M2.5 normal	paired gate	asymptotic diagnostic	Tab. 3
M3	FIT-static label allocation	constructed-mixture descriptive	Sec. 6
M3.5	width-surrogate allocation	diagnostic, not a policy	Sec. 7

Table 1: Method-status taxonomy. “Exact certificate” is conditional on the stated turnover/calibration assumptions; every reported held-out quantity is a dependent diagnostic.

2 PROBLEM AND SETUP

Candidates. A fixed pool of k candidate models $\mathcal{M} = \{M_1, \dots, M_k\}$ trained on a *fit* split, disjoint from all evaluation data.

Subgroups. Data come from G subgroups (we use ground-truth classes). The deployment prior is a weight vector w on the simplex Δ^{G-1} ; per-model mixture risk is $R_i(w) = \sum_g w_g r_{i,g}$ ((1)).

Assumption 1 (subgroup-mix turnover). *During deployment only the mixture weight w shifts; each group’s class-conditional distribution $P(X \mid Y=g)$ is unchanged, so each $r_{i,g}$ is constant. Reward evaluation uses group-IPS reweighting.*

Calibration. A calibration split (disjoint from fit and test) with n_g points in group g . The per-point error indicator of model M_i in group g is a Bernoulli with mean $r_{i,g}$. We get $\hat{p}_{i,g}$, a Clopper–Pearson exact interval $[l_{i,g}, u_{i,g}]$ at cell confidence $\delta_{\text{cell}} = \delta/(Gk)$.

Decision problem. For each deployment mixture w , output either *commit*(i^* , certificate “true regret $\leq \tau$ ”) or *abstain*. Regret of a choice i at w is

$$\text{regret}(i, w) = R_i(w) - \min_j R_j(w),$$

i.e. the loss of choosing i versus the oracle-best model at w . We certify this quantity from calibration data *only*, never touching test.

3 THEORY: LINEAR BANDS AND THE REGRET CERTIFICATE

Throughout, $k = |\mathcal{M}|$, G subgroups, $\delta \in (0, 1)$. Let $\delta_{\text{cell}} = \delta/(Gk)$, and let $[l_{i,g}, u_{i,g}]$ be the exact CP interval for $r_{i,g}$ at confidence δ_{cell} .

Lemma 1 (Linear band joint coverage). *Let E be the event that $r_{i,g} \in [l_{i,g}, u_{i,g}]$ for all (i, g) . By Bonferroni, $\Pr(E) \geq 1 - \delta$. On E , for every $w \in \Delta^{G-1}$ and every i ,*

$$L_i(w) := \sum_g w_g l_{i,g} \leq R_i(w) \leq \sum_g w_g u_{i,g} =: U_i(w).$$

Proof. Each l, u is a CP bound for that cell; Bonferroni over the Gk cells gives $\Pr(E) \geq 1 - \delta$. On E , fix w ; since $w_g \geq 0$, $\sum_g w_g = 1$, termwise monotonicity of the (deterministic) linear combination gives the sandwich. $\square$

Proposition 1 (Sound minimax regret bound (P2)). *Let i^* be any chosen model and define $\text{UB}_{i^*}(w) = U_{i^*}(w) - \min_j L_j(w)$. On E ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{i^*}(w).$$

Proof. $R_{i^*}(w) \leq U_{i^*}(w)$ (Lemma 1), and $\min_j R_j(w) \geq \min_j L_j(w)$ since $R_j \geq L_j$ pointwise. Subtracting the two (weaker) inequalities at their respective endpoints yields the bound; this holds for the comparator j attaining the minimum. $\square$

Proposition 1 is *correct* but, as we show empirically (Sec. 4), unsuitable as a selector. It is a good *gate*: committing i^* whenever $\text{UB}_{i^*}(w) \leq \tau$ certifies true regret $\leq \tau$ with coverage $\geq 1 - \delta$. The tighter the bound, the more mixtures can be committed. This motivates the paired certificate, which is the central finite-sample construction.

Definition 1 (Paired-difference certificate). *For ordered pairs (i, j) , $i \neq j$, define the per-point paired difference on the calibration set*

$$d_{ij}(\mathbf{x}) = \mathbb{1}\{\text{err}_i(\mathbf{x})\} - \mathbb{1}\{\text{err}_j(\mathbf{x})\} \in \{-1, 0, +1\},$$

and its group- g mean $\bar{d}_{ij,g}$, an unbiased estimator of $r_{i,g} - r_{j,g}$. Let $\alpha = \delta/(k(k-1)G)$ (Bonferroni over all ordered pairs $\times$ groups). For each (i, j, g) take a one-sided upper confidence bound $\text{UCB}_{ij,g} \geq r_{i,g} - r_{j,g}$ simultaneously with probability $\geq 1 - \delta$ over all pairs/groups. Candidate variants (Sec. 5, Tab. 3) operate on $X_{ij} = (\bar{d}_{ij} + 1)/2 \in [0, 1]$. The certificate value for chosen i^* at w is

$$\text{UB}_{\text{cf}}(i^*, w) = \max \left\{ 0, \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g} \right\}.$$

Theorem 1 (Paired-difference certificate is sound (MCB-style)). *On the simultaneous event of Definition 1 (probability $\geq 1 - \delta$), for any choice i^* ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{\text{cf}}(i^*, w).$$

Hence if $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$, committing i^* is certified to have true regret $\leq \tau$, with coverage $\geq 1 - \delta$.

Proof. $\text{regret}(i^*, w) = \max\{0, \max_{j \neq i^*} (R_{i^*}(w) - R_j(w))\}$: including the self-comparator contributes deterministically 0, and the minimum is attained by a comparator. If the inner maximum is positive, fix a maximizing $j_* \neq i^*$. Since E holds, $R_{i^*}(w) - R_{j_*}(w) = \sum_g w_g (r_{i^*,g} - r_{j_*,g}) \leq \sum_g w_g \text{UCB}_{i^*j_*,g}$. This sum is exactly one term of the max in UB_{cf} , hence is bounded by it. If the inner maximum is nonpositive, regret is 0 and the explicit outer maximum supplies the required bound. $\square$

The key is *paired* cancellation: UB_{cf} subtracts models evaluated on the *same* calibration points within each group, removing the shared per-point error that can inflate UB_{i^*} in Proposition 1.

Honest scope of the certificate. Each $\text{UCB}_{ij,g}$ is computed once, independently of w (Bonferroni over ordered pairs $\times$ groups only, Def. 1), so the simultaneous event is a single event of probability $\geq 1 - \delta$ under which $\text{UB}_{\text{cf}}(i^*, w) = \max\{0, \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g}\}$ is valid at the same time for *all* w on the simplex and all data-dependent choices $i^*(w)$: the paired certificate is *simultaneous over the continuous simplex*, with no grid-level Bonferroni. We verify this with a dense off-grid Dirichlet w -sample (Sec. 7, 1000 points per (carrier, seed, budget), none on-grid): the dependent held-out diagnostic finds no committed violation (e.g. MNIST $\pi=0.5$ commits 2816/3000 off-grid rows). The certificate certifies *soundness* of commits, not optimal abstention; τ is an operator parameter, and we report the coverage–commitment tradeoff as a fact, not a theorem.

4 WHY BANDS MUST GATE, NOT CHOOSE: M1 VERIFIED NEGATIVE

The most tempting “safety” rule is minimax selection,

$$i_{\text{mmp}}^* = \arg \min_i \left(\max_j \text{UB}_{ij}(w) \right), \quad \text{UB}_{ij}(w) = U_i(w) - L_j(w),$$

i.e. pick the model whose worst-case gap to any other is smallest. We ran it on real data (digits, Fashion-MNIST, MNIST, 20News; $G=\text{class}$; $k=4$; LR/C-variants, SVC, MLP).

Proposition 2 (M1 degeneracy). $\max_j (U_i(w) - L_j(w)) = U_i(w) - \min_j L_j(w)$, so $i_{\text{mmp}}^* = \arg \min_i U_i(w)$: the “max joint” collapses to the minimizer of the upper band, whose lower side ($\min_j L_j$) is constant across i .

When CP widths are homogeneous, $\arg \min_i U_i(w)$ equals the point-estimate argmin (vacuity); under *heteroskedastic* widths the rule is driven by the most pessimistic band and can pick a strictly worse model. On our real grid it matches point-estimate selection on every homogeneous-width row and diverges on mixture-ball vertices without a consistent direction (App. A): the rank opinion is driven by band widths, not by the point estimate—so it neither vacuous-commits nor reliably out-ranks point-estimate selection; bands gate, they do not choose (Prop. 3).

Proposition 3 (Anti-selectivity of M1 under heteroskedastic widths). *For a fixed w , suppose model a has a wide lower-side band driven by a sparse subgroup (so $\min_j L_j(w)$ is small), and the point estimate prefers model b with $\hat{R}_b < R_a$. Then M1 selects a (the pessimistic pick) whenever the width $\Delta_a = U_a - L_a$ exceeds $\hat{R}_b - L_a$; committed regret becomes $R_b - R_a$ (positive) while cal-prior commits b at regret 0 (i.e. M1 is strictly worse whenever the width of the pessimistic model exceeds the true gap).*

Proof. Follows directly from Prop. 2: M1 picks $\arg \min_i U_i(w)$; on the event that $U_a(w) < U_b(w)$ while $\hat{R}_a > \hat{R}_b$, the rule selects the higher-error model, and the “regret” it believes it controls is an artifact of the pessimistic band width, not a real comparison. $\square$

Safety bands are *certificates*, not arrows; we adopt the design they motivate.

5 POINT-ESTIMATE SELECTION WITH CERTIFICATE GATING (M2 / M2.5)

5.1 THE VERIFIED POLICY

Selection is driven by the point estimate, never by the bands:

$$i^*(w) = \arg \min_i \hat{R}_i(w) = \arg \min_i \sum_g w_g \hat{p}_{i,g}.$$

Bands serve *only* as a gate for certificates and honest abstention. We commit $\langle i^*(w), \text{regret} \leq \tau \rangle$ iff $\text{UB}(i^*, w) \leq \tau$ (P2) or $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$ (P3/Thm. 1), and otherwise abstain—falling back to a robust mixture / domain choice rather than a hard pick. This paper does not fix or evaluate that fallback policy or its operator cost; an abstention therefore means only “no certificate at tolerance τ ,” not “better deployment utility.”

Definition 2 (Cert-gated safe selection). *Given tolerance τ and a sound regret upper bound UB:*

$$\text{decision}(w) = \begin{cases} \text{commit } \langle i^*(w), \text{regret} \leq \tau \rangle & \text{UB}(i^*, w) \leq \tau, \\ \text{abstain} & \text{UB}(i^*, w) > \tau. \end{cases}$$

5.2 REAL-DATA VERIFICATION (R1884 M2 + R1885 M2.5)

We verify on four public carriers (digits, Fashion-MNIST, MNIST via PCA-128 frozen features; 20News via tf-idf→PCA-128), classes as subgroups, four classifiers (LR- $C \in \{1, 0.01\}$, LinearSVC, MLP), $\delta=0.1$, and a pre-registered w grid (uniform, collected u , 20 single-class skews, three interpolation peaks). Held-out estimated regret is settled on a test split by group reweighting; test is only ever read. Rates below are *matched* on the same 5 seeds $\{0, 1, 2, 3, 4\}$ (Sec. 9).

Endpoint ($\tau=0.04$)	held-out pass frac.	commit rate	comm. mean/max	abst. mean	forced
M2 (P2, absolute bands)	1.000	0.269	0.0001/0.0019	0.0043	0.0031
M2.5 paired normal (asympt.)	1.000	0.503	0.0009/0.0364	0.0054	0.0031

Table 2: Main endpoints (*matched* 5-seed means, $\tau=0.04$). “1.000” means zero observed held-out estimated-regret violations among committed, dependent grid rows; it is not nominal coverage or a true-regret estimate. The M2 row is exact finite-sample; the paired-normal row is an asymptotic diagnostic. Abstained rows have higher observed forced-pick regret (0.0044 vs. 0.00001), but this does not price a fallback. Same seeds/splits/pool.

Certificate	committed rate	held-out pass frac.	finite sample
M2 absolute CP bands	0.269	1.000	exact
M2.5 Hoeffding (paired)	0.097	1.000	exact
M2.5 Maurer–Pontil (paired)	0.260	1.000	exact
M2.5 normal/CLT (paired)	0.503	1.000	asymptotic

Table 3: **Certificate-price frontier** (matched 5-seed means). The first three rows are exact finite-sample certificates; their reported held-out violation fraction is 0. The normal/CLT row is an *asymptotic diagnostic*: its 0.503 rate and held-out diagnostic cannot support a finite-sample claim. The exact paired Maurer–Pontil bound commits about as many mixtures as M2 ($0.260 \approx 0.269$).

5.2.1 INTERPRETATION

1. **Exact versus asymptotic evidence.** Across the full $\tau \in \{0.02, \dots, 0.10\}$ sweep, the exact M2 and paired exact variants have no observed held-out violations in this snapshot. The normal/CLT variant also has no observed violations, but it is asymptotic and is not used as finite-sample evidence.
2. **Descriptive abstention pattern.** Abstained rows’ forced hard-pick regret is higher than committed rows’ (M2 matched: 0.0044 vs. 0.00001 mean). This descriptive contrast neither selects a fallback nor establishes its utility, latency, or cost.
3. **Per-carrier.** Large-error carriers (digits/20News) stay conservative and honestly abstain; low-error carriers (Fashion/MNIST) commit narrowly-and-safely. Reported heterogeneity, not hidden.
4. **Certificate-price, not rule change.** UB_{cf} cancels the within-CAL shared error that inflates UB_{i^*} in Proposition 1; the chosen i^* is identical, so the raise is honest safety-in-place.

6 LABEL BUDGET AND SUBGROUP ALLOCATION (M3)

The certificates in Sec. 5 use every calibration point. A real deployment of a *rank-question* may come with only a finite *label budget* R : of the unlabeled calibration pool, one may reveal no more than R group labels before having to commit or abstain. How do pre-specified, FIT-only allocations spread those labels across subgroups? We study this restricted question on the same problem and carriers.

6.1 SETUP AND ALLOCATION RULES

Fix $R = \lfloor \pi \cdot N_{\text{cal}} \rfloor$ with $\pi \in \{0.50, 0.65, 0.80, 0.95\}$. All other choices are inherited from M2.5 (point-estimate selection, paired-MPB gate, $\delta=0.1$, 3 seeds, four carriers). Three static rules map $(R, \{\text{per-group avail}\})$ to reveal counts $\{n_g : \sum n_g = R\}$:

- **uniform:** $n_g \propto 1$ (equal across groups).
- **neyman / widthgreedy:** $n_g \propto \hat{\sigma}_g$ (per-group std of the FIT errors across the model pool), i.e. the error-band-width proxy.
- **sens:** $n_g \propto w_g^{\text{worst}} \cdot \hat{\sigma}_g$, where w^{worst} is the mixture vertex with maximal FIT-pool mean risk—sensitivity weighting.

All displayed rules are *pre-specified*: reveal counts are fixed functions of FIT alone, so the static paired-MPB certificate of Thm. 1 remains valid. An earlier adaptive track and its claimed confidence-sequence result are omitted: this snapshot does not specify the observable subgroup information, filtration, sampling model, or low-count treatment needed to establish that theorem for the implemented process.

carrier	budget π	uniform	neyman	sens
MNIST	0.50	0.778	0.400	0.578
MNIST	0.95	1.000	1.000	1.000
Fashion	0.50	0.000	0.067	0.067
Fashion	0.95	0.133	0.111	0.133
digits	0.50~0.95	0.000	0.000	0.000
news	0.50~0.95	0.000	0.000	0.000

Table 4: **M3 committed-rate at $\tau=0.04$, $\delta=0.1$, 3-seed means.** Entries aggregate dependent mixture-grid rows under constructed mixtures. The frozen summary records no held-out estimated-regret violation among the 3360 static-policy cells, but this is a diagnostic, not nominal coverage or an independent uncertainty estimate. No directional allocation comparison is inferential here.

6.2 FINDINGS

1. **Static-gate scope.** For FIT-pre-specified allocations, Thm. 1 applies under its stated assumptions. The snapshot’s 0/3360 held-out violation count is a dependent settlement diagnostic, not an estimate of nominal coverage.
2. **Allocation contrasts are exploratory.** Uniform is 0.778 versus 0.400 (neyman/widthgreedy) and 0.578 (sens) on the MNIST $\pi=0.5$ summary; Fashion reverses the displayed ordering at low budget. Frozen M3 records contain only three-seed aggregate rows, not per-seed outcomes, so these comparisons are descriptive rather than supported directional conclusions.
3. **Missing defer costs.** We record forced hard-pick and FIT-anchor regret diagnostics (up to ~ 0.055 digits / ~ 0.048 news), but neither is a precommitted fallback policy and their utility, latency, and monetary costs are unmeasured. Details and the honest missing-data/sensitivity protocol are in App. D.

7 FIXED-BUDGET CERTIFICATE-WIDTH SURROGATE

The M3 experiments (Sec. 6) reported an empirical law—uniform spread out-commits variance-aware rules on the strongest carrier—but left its *mechanism* implicit. This section studies a stylized *certificate-width surrogate* to help interpret those results. It does **not** optimize the actual candidate-indexed paired regret UCB $\max_j \sum_g w_g \text{UCB}_{i^*j,g}$, establish a safe allocation theorem, or turn a commit-rate comparison into a deployment allocation recommendation.

7.1 PROBLEM AND WATER-FILLING SOLUTION

Let the budget R be split into per-group reveal counts n_g , $\sum_g n_g = R$. The actual paired certificate also contains candidate-specific empirical means, variances, bias terms, and a maximum over comparators. We deliberately discard all of these and retain only the stylized half-width $r_g(n) = \beta_g / \sqrt{n}$, where β_g is a FIT-visible dispersion proxy. For each fixed mixture w write $c_g(w) = w_g \beta_g$ and use the extended-value convention $c_g / \sqrt{0} = 0$ when $c_g = 0$ and $c_g / \sqrt{0} = +\infty$ when $c_g > 0$. The resulting objective is a width diagnostic, not a bound on regret or on any actual certificate:

$$\min_{n_g \geq 0, \sum_g n_g = R} \max_{w \in W} \sum_g w_g r_g(n_g). \quad (\text{WS})$$

Equation (WS) is the *multi-mixture* surrogate. The following closed form is only for its one-mixture subproblem $W = \{w\}$: with $A = \{g : c_g(w) > 0\}$, the continuous relaxation puts $n_g^* = 0$ for $g \notin A$ and, for $g \in A$, obeys equi-marginality:

$$\forall g \in A : \frac{1}{2} n_g^{-3/2} c_g(w) = \lambda \implies n_g^* = R \frac{c_g(w)^{2/3}}{\sum_{h \in A} c_h(w)^{2/3}}, \quad (2)$$

where λ is the budget multiplier. If $A = \emptyset$, the one-mixture surrogate objective is identically zero and every allocation is optimal. The $2/3$ exponent comes from the $\beta / \sqrt{n}$ scale (error-indicator concentration); it changes to $1/2$ or 1 for log or β/n costs. A bisection fill solves the budget equation in $O(G \log R)$.

Proposition 4 (Uniform optimality for one fixed width surrogate). *For a single fixed mixture $W = \{w\}$ with at least one positive coefficient $c_g = w_g \beta_g$, the all- G -group uniform allocation $n_g = R/G$ is the unique minimizer of the width surrogate (WS) iff $c_g = c > 0$ for every group g . If some $c_g = 0$ and some $c_h > 0$, the minimizer is on the boundary ($n_g^* = 0$ for zero-coefficient groups), so the all- G -group uniform allocation is strictly suboptimal. If every $c_g = 0$, the objective is identically zero and uniform is merely one of many minimizers.*

Sketch. When $A \neq \emptyset$, (2) is the unique optimum on its positive support. It equals R/G in every coordinate exactly when A is all groups and all c_g are the same strictly positive value. If A is a strict subset, moving any budget from $g \notin A$ to a positive-coefficient group strictly lowers the objective. The $A = \emptyset$ case follows from the convention above. $\square$

Multi-mixture surrogate calculation. For a finite predeclared $W = \{w_1, \dots, w_K\}$, the optimum is *not* obtained by applying the preceding formula to an arbitrarily chosen binding mixture. Writing each predeclared mixture as an engine with coefficient $C_{kg} = w_{k,g} \beta_g$, the surrogate uses the diagnostic dual:

$$\begin{aligned} d^* = P^* &= \min_n \max_{\mu \in \Delta^K} \sum_{k,g} \mu_k C_{kg} n_g^{-1/2} \\ &= R^{-1/2} \left[\max_{\mu \in \Delta^K} S(\mu) \right]^{3/2}, \quad S(\mu) = \sum_g \left(\sum_k \mu_k C_{kg} \right)^{2/3}, \end{aligned} \quad (3)$$

a convex diagnostic program (with S concave on Δ^K) solved by projected-gradient ascent. Complementarity identifies binding *surrogate mixtures*, not actual candidate comparators: 6–25% bind in these calculations, and the all-active heuristic changes the surrogate value by 1.17–4.19 $\times$ (App. B). On digits/news the surrogate stays above τ at every $\pi \leq 0.95$; this is a width-wall diagnostic, not infeasibility of the actual certificate.

7.2 DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

For the one-mixture surrogate, a deterministic construction (App. C) shows that uniform need not minimize stylized width: on an asymmetric non-spanning $W_0 = \{e_0\}$, three coefficients are zero, so continuous water-filling puts the *surrogate* budget on the queried group. This is a calculation about Eq. (WS), not a statement that the resulting allocation lowers the actual paired regret UCB or commits safely.

7.3 SAME-BUDGET ALLOCATION COMPARISON

We add a surrogate-water-fill allocation (with β_g from FIT-only cross-model paired-diff std) as a descriptive same-budget comparator. The paired-MPB certificate is evaluated anew for every allocation; the surrogate itself neither upper-bounds it nor selects a safe allocation. Table 5 reports committed-rate at $\tau=0.04$, 3-seed means.

carrier	budget	uniform	neyman	width	sens	width-surrogate
MNIST	0.50	0.778	0.400	0.400	0.578	0.400
MNIST	0.80	1.000	0.844	0.844	1.000	0.889
Fashion	0.50	0.000	0.067	0.067	0.067	0.067
Fashion	0.80	0.067	0.089	0.089	0.067	0.089
digits/news	all	0.000	0.000	0.000	0.000	0.000

Table 5: **M3.5 width-surrogate comparator**, $\tau=0.04$, $\delta=0.1$, **3-seed means**. Same grid and paired certificate as M3; the frozen dependent held-out diagnostic has no committed estimated-regret violation. These are descriptive paired-certificate outcomes, not actual regret-UCB optimality: uniform leads on MNIST, the surrogate ties neyman/width on low-budget Fashion, and digits/news never commit.

Post-hoc diagnostic slices. $CV(\hat{\beta})$ and spanning remain descriptive, not conditional allocation gates. The detailed $W' = \{e_{g_0}\}$ slice and the synthetic characteristic are retained only in the appendix because neither validates the actual regret-UCB objective nor creates a policy.

8 RELATED WORK

Model selection under shift. Group/DRO objectives (Sagawa et al., 2019; Duchi & Namkoong, 2019) optimize a predictor objective rather than this paper’s output—a candidate-pool choice or abstention with mixture-indexed regret certificate. Prior adjustment (Saerens et al., 2002) recalibrates a chosen classifier; it does not supply the present continuous-mixture candidate-ranking certificate. These are scope contrasts, not priority claims.

Finite-sample certificates. We apply Clopper–Pearson and simultaneous-confidence ideas (Vovk et al., 2005; Clopper & Pearson, 1934) to selected-model, mixture-indexed regret. The exact paired bound uses Maurer–Pontil empirical Bernstein (Maurer & Pontil, 2009) and best-vs-rest paired comparisons (Dunnett, 1955; Hsu, 1984); our contribution is restricted to their instantiated decision object, assumptions, and certificate.

Label budgeting (M3). Unlike per-point audit or importance re-weighting for a fixed model, M3 allocates calibration reveals across subgroups while certifying mixture-indexed regret over candidates; it is not a per-point value estimator.

9 LIMITATIONS

- **Finite-sample vs. asymptotic.** The paired certificate is simultaneous over the continuous simplex; the off-grid scan has no observed held-out violations. Only Maurer–Pontil/Hoeffding are finite-sample; normal/CLT is asymptotic. The exact relative gate has only trivial status-quo commits, so strict operators should use the absolute gate.
- **τ , dependence, and seeds.** We report a certificate-gate/commitment tradeoff, not an optimal- τ claim. M2/M2.5 use matched five seeds; M3/M3.5 use three-seed aggregate budget means without retained per-seed allocation records, so allocation directions are exploratory.
- **Turnover model.** Assumption 1 (fixed class-conditional) does not cover covariate shift inside a group; a distinct failure mode we do not certify.

10 CONCLUSION

Under the stated turnover and calibration assumptions, exact CP and paired bounds certify point-estimate ranking through a regret gate; the paired construction is simultaneous over the continuous simplex and its frozen dependent held-out diagnostic has no committed violations. Normal/CLT is asymptotic only. The exact absolute Maurer–Pontil gate retains content (0.260 committed), whereas the exact relative gate admits only trivial status-quo commits; its informative switches are asymptotic diagnostics. The allocation analysis remains a width surrogate, not an actual regret-UCB policy, and natural time/geo shift plus fallback cost remain open.

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A M1 DIAGNOSTIC DETAIL

On the flat mixture grid, minimax-upper-band ($M1$) is identical to point-estimate selection on 350/350 rows (vacuity: homogeneous widths collapse $\arg \min_i U_i$ to the point-estimate $\arg \min$). On mixture-ball vertices that hedge over an uncertainty set it diverges on 165/352 rows but with no consistent direction on the diverged subset (mean regret diff -0.0021 : $M1$ slightly *better* on 93 rows, worse on 62), while over all 352 vertices it is slightly worse overall ($+0.0019$). On a diagnostic unbounded grid it diverges in 15/880 rows; those rows are mostly tied (11 of 15), the remaining 4 strictly worse for $M1$ (0.0024 vs. 0.0).

Setting	rows	diverge	regret diff
Flat grid (uniform/skew/interp)	350	0/350	$\equiv$ cal-prior
Mixture-ball	352	165/352	-0.0021 div. (93/62), $+0.0019$ all
Diagnostic grid	880	15/880	0 (11 tied; 4 worse for $M1$, diff $+0.0024$)

Table 6: Minimax-upper-band ($M1$) vs. point-estimate selection. “diverge” = rows where $M1$ ’s choice differs from cal-prior; regret diff = $M1$ regret minus cal-prior regret (positive = worse).

B ALL-ACTIVE WIDTH-SURROGATE ABLATION

For the multi-mixture width surrogate only, complementary slackness identifies a small fraction of binding engines. The dual value is not the actual candidate/mix regret UCB; the all-active column reports a ratio between two surrogate calculations.

carrier	$ K $	active engines	$d_{\text{mm}}^*/d_{\text{allactive}}^*$
MNIST	60	25%	1.30
Fashion	60	25%	1.33
digits	60	7%	4.19
news	100	20%	1.17

Table 7: All-active heuristic ablation for the width surrogate (Eq. 3), not for the actual paired-regret UCB.

C DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

Take $G=4$, $R=1200$, per-group paired-error std $\beta = (0.10, 0.10, 0.02, 0.02)$, true per-group gaps $\delta = (0.02, 0, 0.02, 0)$, and an *asymmetric* (non-spanning) deployed set $W_0 = \{e_0\}$ (only subgroup 0 is ever queried):

rule	n_g	reported paired UCB at e_0	actual commit
uniform	(300, 300, 300, 300)	0.094	no
width-surrogate	(1197, 1, 1, 1)	0.029	yes

Here $c = (0.10, 0, 0, 0)$, so Prop. 4 gives the continuous surrogate optimum $(R, 0, 0, 0)$; uniform is strictly suboptimal because it wastes labels on the three zero-coefficient groups. The displayed integer implementation enforces one label per inactive group and therefore uses (1197, 1, 1, 1), a near-boundary realization. Its reported paired UCB and “actual commit” are separate paired-certificate outputs, not numerical evaluations of Eq. (WS) and not an allocation guarantee for the actual UCB.

D DEFER COSTS IN M3

The snapshot records forced point-estimate and FIT-anchor regret diagnostics, but neither fallback was precommitted as an operational policy and no latency, monetary, switch, or abstention penalty

was measured. Thus the reported hard-pick regret (up to ~ 0.055 digits / ~ 0.048 news) is not a fallback-cost estimate. The planned real-shift protocol records these fields as missing or conducts an explicit sensitivity analysis; no utility conclusion is made here.

E GATE OPERATING CHARACTERISTIC

deployed set	$CV(\hat{\beta})$	signed area = $\text{mean}_{\tau}(\text{rate}_{\text{surrogate}} - \text{rate}_{\text{unif}})$	
		area	worst τ gap
non-spanning W'	low (sym.)	+0.25	—
non-spanning W'	high	+0.25	—
spanning W	low (sym.)	+0.00	0
spanning W	high	-0.08	-0.49 at $\tau=0.12$

Table 8: Diagnostic operating characteristic: committed-rate difference width-surrogate – uniform vs. τ ($G=4$, $R=1200$, 20 seeds, same paired-MPB certificate). This describes a synthetic comparison only; it is not a safe-allocation gate or an optimization result for the actual regret UCB. Evidence: `results/SUBGMIX_GATE_OC_R1899.json`.

F CHOOSING OR ELIMINATING THE TOLERANCE τ

The certificate band is τ -agnostic: τ enters only the comparison $D(i^*, w) \leq \tau$, never its construction (Sec. 3, Prop. 1). We exploit this to close the “ τ -selection protocol” gap in two complementary ways ($M6$, $M7$), with the same calibration fraction, delete budget, and paired normal band. $M7$ runs out-of-distribution from the main evidence on an unseen seed set $\{5, 6, 7, 8\}$ (the main tables use seeds $\{0, \dots, 4\}$); $M6$ runs on the same seeds $\{0, \dots, 4\}$ as the main tables, and we additionally report its strictly-finite-sample band behavior below.

$M6$: eliminate τ by shifting the control object. Instead of certifying regret $\leq \tau$ against an oracle-best model, certify the deployment decision against the *status-quo model the operator would have used anyway*: F_0 = point-estimate-optimal collected-mixture model (no certificate needed to run it). The gate proposes the certificate-selected $i^*(w)$ only if a one-sided paired UCB shows it is not worse than F_0 , else keeps F_0 . This is a *relative* certificate (“no worse than what I would otherwise run”), and τ never appears. Across all $350 \text{ rows} \times 4 \text{ carriers}$, the upgrade rate is 0.663; the squared regret difference $R_{\text{dec}} - R_{F_0}$ has mean -0.0015 and *maximum* 0.0, i.e. no-worse-than- F_0 held-out diagnostic violation fraction 0. On the 118 rows where $M6$ abstains, running the naive always-commit rule instead would make 23 (19%) genuinely worse (max additive regret $+0.028$); $M6$ ’s max regret is 0 there. This is a different, *complementary* statement to the absolute regret $\leq \tau$ certificate: an operator who needs an absolute bound uses the frozen $M2.5$; an operator who wants a safe upgrade over a fixed incumbent uses $M6$.

$M7$: keep absolute τ , select it from a finite menu on calibration only. Take a finite menu $T = \{0.01, \dots, 0.05\}$ and the prespecified rule $\hat{\tau} = \min\{\tau \in T : \text{CR}_{\text{cal}}(\tau) \geq 0.5\}$ (else the tightest menu entry), using only the calibration block’s group errors — never the outer/test block. Because the band holds simultaneously independent of τ , the resulting $\hat{\tau}$ carries *zero additional certificate multiplicity*: every committed point still satisfies true regret $\leq \hat{\tau}$ on the same $\geq 1 - \delta$ event. The price of selection is confined to the performance layer. On OUTER settlement, CAL-select yields committed rate 0.357 with no observed held-out violation (mean committed regret 0.0003, max 0.0053), picking conservatively tight menu entries ($\hat{\tau}=0.01$ on 12 of 16 carrier $\times$ seed blocks). A test-snooping variant that picks τ on OUTER and reports the same test inflates committed rate to 0.539 ($+0.18$) with a 0.7% held-out estimated-regret violation rate — the precise price of the data-peeking the CAL-only rule avoids. Weak domains are reported unfiltered: digits commits 0/60 rows (capacity wall) and news 4/100 (budget wall).

Strictly finite-sample bands in the relative gate ($M6$ continued). The main $M6/M7$ findings use the *asymptotic* paired normal band (tightest). Because the overall conclusion names a strictly finite-sample (non-asymptotic) tightened band as its only remaining item—already closed for the

absolute gate by the Hoeffding/Maurer–Pontil columns of Tab. 3—we close the same item for the *relative* $M6$ gate by recomputing its one-sided paired difference band at the exact finite-sample Hoeffding and Maurer–Pontil widths (same splits, same Bonferroni split, same F_0 and i^* ; base fields reproduce the frozen $M6$ FILE EXACTLY under the normal band). Under strict finite-sample rigor, committed rate drops modestly ($0.663 \rightarrow 0.646$, matched 5-seed) and the no-worse-than- F_0 held-out diagnostic still has zero violations. But the mechanism is a *vacuity collapse*: all 226 rows with $i^*=F_0$ commit trivially (regret difference 0), while the *only* informative rows—the 6 where a candidate $i^* \neq F_0$ is proposed as an upgrade—are all *rejected* by the exact bands (empirical paired difference $D \in [-0.036, 0.098]$, width ~ 0.25); D is never negative enough for the exact width to certify a switch, so no strictly-finite-sample commit ever differs from the status-quo. Equivalently: the relative gate’s only non-trivial content is carried by the asymptotic normal band; strict finite-sample rigor makes it empty. This is a *boundary characterization*, not a defect: if strict rigor is required, the operator should use the *absolute* gate (M2.5), whose exact Maurer–Pontil band retains genuine content (0.260 committed, with zero held-out estimated-regret violations). The relative no-worse-than- F_0 statement is meaningful exactly because it is allowed the asymptotic band; narrowing the band removes the only reason to prefer it over the absolute certificate. This collapse is, however, *cheap to reverse*: measuring the fixed empirical per-group paired differences under an effective calibration-sample multiplier N (same realized stats, band width scaled as $n_g \rightarrow Nn_g$), all 6 switch proposals reopen under the exact Maurer–Pontil band at $N^* \in \{2, \dots, 10\}$ (median 5) and under Hoeffding at $N^* \in \{2, \dots, 20\}$ (median 10), with OUTER soundness intact at every opening point (oracle switch gain $0.034\text{--}0.130 > 0$); monotone in N , so no opening admits a regret-increasing switch. Thus strict finite-sample rigor on the *relative* gate is not infeasible but *budget-priced*: a modest effective 5–20 $\times$ larger calibration set restores the non-trivial content that the asymptotic band currently carries, and the frontier is per-mixture measurable at certificate time.

$M9/M10'$ (audited): exact-band non-emptiness is a provable function of the label budget, not merely observed. On fresh seeds $\{10, \dots, 14\}$, disjoint from both the frozen certificate seeds $\{0, \dots, 4\}$ and the $M7$ calibration seeds $\{5, \dots, 8\}$, we pre-fix a per-group calibration-label budget grid $b \in \{0.25, 0.5, 1.0\}$ and, at each schedule, *actually* subsample each group’s calibration block, re-select F_0/i^* /misclassification, and recompute every band empirically. Writing the selector margin $\Delta(w) = \hat{R}_{F_0}(w) - \hat{R}_{i^*}(w) \geq 0$, any strictly finite-sample band satisfies $\text{UCB}_g \geq \mu_g$, so admission $\sum_g w_g \text{UCB}_g \leq 0$ forces $\sum_g w_g \text{bw}_g(n_g) \leq \Delta(w)$. Under proportional per-group budgets $n_g = b n_g^{\text{full}}$ and the Hoeffding width $\text{bw}_g = c/\sqrt{n_g}$, a *necessary* budget is $b \geq b^*(w) := \lceil c \sum_g w_g / \sqrt{n_g^{\text{full}} / \Delta(w)} \rceil^2$. When $b^* > 1$ —as it does for *all* 125 real-switch rows ($i^* \neq F_0$) at the full split across the fresh seed set (median 273, min 3.7; Δ median 0.014)—no feasible budget $b \leq 1$ admits one, a data-specific *provable* emptiness certificate over the whole feasible axis (the empirical contrast: exact Hoeffding/Maurer–Pontil bands admit 0 real switches at every budget on every carrier). The derivation above isolates the *proportional* allocation $n_g = b n_g^{\text{full}}$; the whole-axis statement is not a free extrapolation. It closes over *all* relaxations of the per-group labels as follows: any feasible allocation has $n_g \leq n_g^{\text{full}}$, and the Hoeffding width $c/\sqrt{n_g}$ is monotone-decreasing in n_g , so $\sum_g w_g \text{bw}_g(n_g) \geq \sum_g w_g \text{bw}_g(n_g^{\text{full}})$ —the admission-weighted bandwidth is minimized at the all-caps corner, i.e. the full proportional allocation itself. Since that corner already rejects (its MPB admission $\sum_g w_g \text{bw}_g^{\text{MPB}}(n_g^{\text{full}}) > 0$ on all 125 rows), reallocating labels toward any subgroup or pair of subgroups—including the M3 water-filling move applied to the *relative* gate—can only *raise* the weighted bandwidth and leave every row rejected. Emptiness is monotone in the allocation as well as in the budget, so the certificate holds over the full {allocation $\times$ budget ≤ 1 } box, not just its proportional slice (verified against the frozen rows; the sparse MPB Bernstein bias $7L/(3(n-1))$ is non-monotone only below every realized per-group cap ≥ 39 , so MPB also closes the real box). Strictly separating trivial submissions ($i^*=F_0$, regret difference 0) from real switch proposals throughout, trivial rows never inflate any reported rate. The *absolute* exact gate (M2.5) is the deployable safety rule and retains content on the same fresh seeds, monotonically in budget (fashion committed rate $0.49 \rightarrow 0.96 \rightarrow 1.0$ as b grows, with zero held-out no-worse-than- F_0 violations; mnist commits at rate 1.0 at every budget), so the operator rule is: deploy under the exact *absolute* gate, treat the relative gate as an asymptotic/descriptive diagnostic only. Weak domains survive honest reporting—news is a *budget* wall (committed rate 0.136 at $b=1$, 0 at $b=0.25$; its sole $b=0.5$ commit is not no-worse), digits a *capacity* wall (0.093).

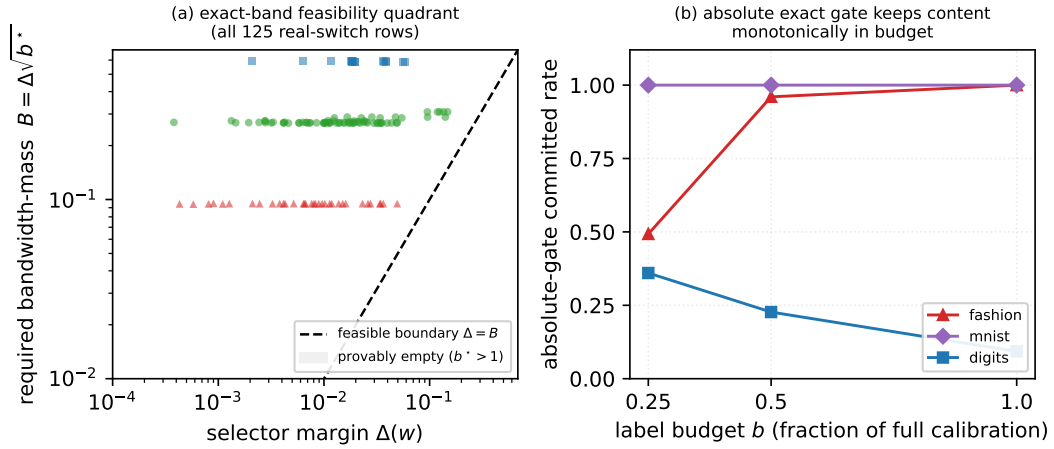


Figure 1: **Certificate-time emptiness is a provable function of the label budget.** (a) The 125 real-switch rows ($i^* \neq F_0$) at the full split across the fresh seeds $\{10, \dots, 14\}$, in the *selector-margin* $\Delta(w)$ vs. *required bandwidth-mass* $B = \Delta \sqrt{b^*(w)}$ quadrant (log-log; marker keyed by carrier). Admission under any strictly finite-sample band forces $\Delta(w) \geq B$ (the feasible half-plane below the dashed $\Delta=B$ line); every realized row lies strictly above it, i.e. its certified emptiness $b^*(w) > 1$ is *provable* over the whole feasible axis, not merely observed. (b) The deployable *absolute* exact gate (M2.5) instead retains content monotonically in the label budget b : committed rate grows with b on fashion/digits while mnist commits at every budget, all with zero observed held-out no-worse-than- F_0 violations (Sec. F). The relative gate (M6) is an asymptotic/descriptive diagnostic only.

M12 (fresh-seed audit of the M3/M3.5 budget totals). The M10/M11 emptiness certificates fix a per-group *label budget* $n_g = b n_g^{\text{full}}$ on fresh seeds $\{10, \dots, 14\}$. Here we re-drive the same object at the M3/M3.5 *total* budgets $R = \lfloor \pi N_{\text{cal}} \rfloor$, $\pi \in \{0.5, 0.65, 0.8, 0.95\}$, by *real* without- replacement *FIT/CAL/OUTER* sampling (same four carriers, same pool, same $\delta, d_{\text{cell}}, w$ -grid, paired bands), re-splitting R among subgroups with the pre-specified M3 rules (uniform/neyman/sens/width-greedy) and the M3.5 width-surrogate water-fill, all reading *FIT* only so the static paired-MPB certificate stays valid. Across all 100 cells (4 carriers $\times$ 4 budgets $\times$ 5 allocations $\times$ 5 fresh seeds), the exact *relative* band (Hoeffding or MPB) admits 0 **genuine switches** ($i^* \neq F_0$ committed): every reported exact-relative “commit” is a trivial keep-status-quo row, so the M10/M11 whole-axis emptiness persists unchanged at the M3 totals by real sampling, not extrapolation. The exact *absolute* gate (M2.5) has no held-out estimated-regret violation on the 37 committing cells, with content monotone in the budget (MNIST uniform $0.773 \rightarrow 1.0$; Fashion $0 \rightarrow 0.20$ at $\pi=0.95$, switch gain mean 0.0131/max 0.0753, worst harm -0.0012), and the weak domains keep their walls (digits/news 0 committed everywhere). Evidence: subgroup_mix_ranking/results/SUBGMIX_M12_FRESH5_BUDGET_R1915.json.

G AI USE STATEMENT

Large language models, including Codex, assisted with implementation support, proof and cross-reference auditing, bibliography and metadata checking, manuscript revision, and internal review workflow support. They did not generate experimental observations, choose unreported results, or replace author responsibility for scientific claims, experiment design, interpretation, or the final manuscript. This statement includes the present revision and audit cycle.